RACoN

Research Validation & EB-2 NIW Evidence Development Plan

Team Meeting Project Paper | Version 1.0 | October 27, 2026

Metademic Research Initiative • Resource-Aware Collaborative Network

PROJECT REFERENCES	WORKING DOCUMENT
Website	https://racon.metademic.org
Source repository	https://github.com/metademicteam/RACoN
Meeting date / participants	_____
Version / commit hash	_____
Prepared for	Development team, research team, project lead, immigration counsel

Purpose: turn RACoN technical claims into independently reviewable evidence, a reproducible research record, and a carefully supported proposed-endeavor narrative for an EB-2 National Interest Waiver petition. This is a planning document, not a certification of functionality or a legal opinion.

Evidence rule: distinguish IMPLEMENTED, TESTED, IN PROGRESS, PLANNED, and NOT VERIFIED. Do not describe a planned capability as a demonstrated result.

1 | Executive Project Brief

RACoN is described by its project lead as a decentralized, resource-aware system in which the requesting machine acts as a temporary coordinator, discovers available resources from other participating machines, distributes portions of AI inference work, and returns the combined output. The team reports tests involving models such as Gemma 3 and Dolphin. These are team-reported claims pending documented verification.

Primary technical question: has a single AI model inference been partitioned and executed across multiple heterogeneous machines, with verifiable per-node computation and correct end-to-end outputs?

Primary immigration question: can the project demonstrate substantial merit and national importance, the petitioner's positioning to advance it, and a persuasive overall basis for waiving a job offer and labor certification under Matter of Dhanasar?

2 | Success Criteria & Decision Gates

Gate	Evidence required	Decision
G1 - Architecture truth	Code map, diagram, coordinator protocol, model partition method	Go / revise claims
G2 - Reproducible execution	Model version, 5-node logs, per-node workloads, successful repeated inference	Go / debug
G3 - Measured benefit	Controlled baselines, latency, throughput, memory, network overhead, cost	Go / qualify
G4 - Research novelty	Literature matrix, comparison with prior art, limitations	Go / refine
G5 - NIW dossier	Dated exhibits, role evidence, external interest, roadmap, counsel review	File-ready / gaps

3 | Twelve-Week Work Plan

Weeks	Workstream	Deliverables / accountable lead
1-2	Architecture and audit	System map, repo inventory, exact claims ledger / Tech lead
3-4	Reproducibility	Environment manifest, model hashes, automated scripts / ML lead
5-6	Benchmarks	Single-node and multi-node trials, raw CSV and logs / QA lead
7-8	Reliability and security	Fault injection, security assessment, limitations / Systems lead
9-10	Research comparison	Related-work matrix, experimental paper draft / Research lead
11-12	Impact and petition evidence	U.S. use cases, external validation plan, exhibit index / PI + counsel

4 | Meeting Operating System

Weekly 60-90 minute meeting: (1) 10 minutes status and blockers; (2) 20 minutes demonstration or raw evidence review; (3) 25 minutes unresolved questionnaire items; (4) 15 minutes risk decisions; (5) 10 minutes assign owners, due dates, and next experiment.

Each question receives: status, owner, answer, proof link, verification reviewer, due date, and decision. Use a shared tracker; no verbal-only closure.

5 | Complete Team Questionnaire (100 Questions)

A. System architecture and innovation

1. What is the complete RACoN architecture?
2. Is the network truly decentralized, and which components remain centralized?
3. How does it operate without a permanent central server?
4. How are new nodes discovered and authenticated?
5. How is the requesting node established as temporary coordinator?
6. What happens when several nodes submit inference requests simultaneously?
7. How does the system respond to node failure?
8. How is the architecture different from existing distributed AI frameworks?

B. AI model partitioning

9. Does RACoN partition actual model layers or computation graphs?
10. Is partitioning layer-, tensor-, pipeline-, or another form of parallelism?
11. What algorithm determines the partitions?
12. How are RAM, VRAM, CPU, and GPU capacity considered?
13. Does partition size adapt automatically to heterogeneous hardware?
14. Are model weights actually distributed across nodes?
15. Does every assigned node execute a real portion of inference?
16. How are intermediate activations or tensors transferred?
17. How are computation and synchronization coordinated?
18. How is the final inference result reconstructed?
19. What happens when a node lacks sufficient memory?
20. Which logs prove actual cross-node model execution?

C. AI model tests

21. Which exact AI models have been tested?
22. Which Gemma 3 variant was used (1B, 4B, 12B, 27B, etc.)?
23. Which Dolphin model and release were tested?
24. What are the model sizes, formats, and quantization settings?
25. How many computers participated in each experiment?
26. Did five computers compute on the same inference request?
27. What was the per-node CPU, RAM, GPU, and VRAM usage?
28. What machine-level evidence proves distributed execution?
29. Can RACoN run a model that the initiating machine cannot run alone?
30. Are results repeatable on the same hardware and workload?

D. Performance and benchmarking

31. What are single-node versus RACoN end-to-end latencies?

32. What are tokens per second in each configuration?
33. What is time-to-first-token?
34. How does performance change at 2, 3, 4, and 5 nodes?
35. How much network overhead is introduced?
36. How long does model loading take?
37. How does total memory use change across nodes?
38. How much network traffic is generated per inference?
39. How do CPU-only and GPU-assisted setups compare?
40. When, if ever, does RACoN outperform the single-node baseline?
41. What controls make the baselines fair?
42. What are the run-to-run variance and confidence intervals?

E. Resource-aware scheduler

43. Which scheduling algorithm is implemented?
44. How are available hardware resources measured?
45. How often is resource availability refreshed?
46. Are bandwidth and network latency part of placement decisions?
47. What happens when a node becomes overloaded?
48. Can an active partition be migrated to another node?
49. Are scheduler decisions logged and reproducible?
50. How does scheduling compare with baseline algorithms?

F. Reliability, networking, security

51. Does RACoN work over both LAN and remote connections?
52. What roles do Tailscale and other networking components play?
53. Can inference recover from a disconnected node?
54. How is data integrity protected during interruptions?
55. Are connections encrypted and how are keys managed?
56. How are unauthorized nodes prevented from joining?
57. How are model weights and user prompts protected?
58. How are malicious or compromised nodes detected?
59. How is user consent obtained for resource sharing?
60. What are current scalability and security limitations?

G. Software and repository evidence

61. Which components are fully implemented in the repository?
62. Which features are simulated, experimental, or incomplete?
63. Does the README accurately describe current behavior?
64. What functionality do automated tests actually cover?
65. Are there end-to-end distributed-inference integration tests?

66. Does the installer work on a clean Windows installation?
67. How are releases, tags, and commit history preserved?
68. Who contributed each major component, and how is authorship documented?
69. What is the software license and IP ownership status?
70. Can an independent team reproduce the tests from the documentation?

H. Scientific novelty and publication

71. What exact research problem does RACoN address?
72. What documented gap exists in the literature?
73. How does RACoN compare with Petals, Ray, DeepSpeed, and relevant llama.cpp approaches?
74. Which contributions are demonstrably novel?
75. Which algorithms are reused versus independently developed?
76. What is the falsifiable research hypothesis?
77. Which experiments test that hypothesis?
78. Which public workloads or datasets support comparison?
79. What negative results and limitations must be disclosed?
80. What is missing for a peer-reviewed manuscript?

I. U.S. national importance

81. Which specific U.S. technological problem could RACoN address?
82. How might small laboratories, universities, or SMEs benefit?
83. Can reliance on expensive AI infrastructure be reduced?
84. How would any cost reduction be measured credibly?
85. Could benefits extend beyond one employer or client?
86. Which U.S. sectors have realistic adoption use cases?
87. Is there documented interest from independent organizations?
88. Are external pilots or evaluations feasible?
89. What are the one- and three-year deployment milestones?
90. What U.S. partnerships, funding, and resources are required?

J. Researcher and NIW evidence

91. What were the principal investigator’s specific original contributions?
92. Which publications directly support this proposed endeavor?
93. Which citations relate to the underlying technical field?
94. What independent recognition or third-party use can be documented?
95. How do memberships, reviews, and credentials relate to the endeavor?
96. Are there genuine patent filings or invention disclosures tied to RACoN?
97. Can independent experts give specific evidence-based assessments?
98. How does prior experience position the petitioner to advance this work?
99. Which records support a realistic future research and execution plan?

100. Why would waiving the job offer and labor certification benefit the United States overall?

6 | Evidence Collection Protocol

For every experiment capture: experiment ID; UTC/local timestamp; git commit; model name/version/hash; quantization; prompt/workload; seed and decoding settings; node IDs; CPU/GPU/RAM/VRAM; OS/drivers; network topology; bandwidth/RTT; placement decisions; per-node compute logs; outputs; failures; and repeated-run metrics.

Evidence ID	Description	Owner	Verification	File / URL
EV-001	Architecture + dataflow diagram	_____	_____	_____
EV-002	Five-node hardware manifest	_____	_____	_____
EV-003	Model-partition execution trace	_____	_____	_____
EV-004	Raw benchmark data + scripts	_____	_____	_____
EV-005	Independent reproduction report	_____	_____	_____
EV-006	U.S. use-case / external interest	_____	_____	_____

7 | Benchmark Matrix

Trial	Nodes	Model / format	Baseline	Measures
B1	1	Gemma 3, exact variant	Native single-node	Latency, tok/s, peak RAM
B2	2	Same model, same settings	B1	Per-node compute + network
B3	3	Same model, same settings	B1	Latency, tok/s, failure rate
B4	4	Same model, same settings	B1	Resource efficiency
B5	5	Same model, same settings	B1	Output correctness + overhead
B6	1-5	Dolphin, exact variant	Comparable implementation	All above + repeatability

Use at least three repeat runs per configuration as a starting point; increase repetitions if variability is high. Report both successful and failed runs. Never claim faster, cheaper, or more efficient without measured, fairly controlled comparisons.

8 | NIW Evidence Map

Dhanasar element	Potential RACoN exhibits	Evidence gap to close
Substantial merit and national importance	Research plan, broader U.S. use cases, literature, pilots, independent letters	Show prospective impact beyond one company or isolated project
Well positioned to advance	Code history, reproducible prototype, papers, citations, expertise, prior achievements	Prove personal contribution and credible progress
On balance, waiver beneficial	Specific future work, collaboration feasibility, national benefits, attorney analysis	Explain waiver rationale without unsupported claims

9 | Risk & Claim-Control Register

Risk	Mitigation
Model is only task-routed, not actually partitioned	Use precise description; do not call it model-parallel until traces prove it.
Claims of cost or speed savings are unmeasured	Use “potential” and publish controlled benchmark results.
Repository differs from marketing site	Audit release tags and correct unsupported website statements.
Founder / nonprofit status overstated	Describe Metademic as an independent research initiative unless registered status is documented.
F-1 unauthorized work exposure	Review activities and authorization with DSO and qualified immigration counsel.
Weak national-importance nexus	Develop external U.S. use cases and third-party evidence; avoid generic AI claims.

10 | Meeting Minutes & Action Tracker

Question / action	Owner	Due	Evidence link	Status
Confirm model partition mechanism	_____	_____	_____	_____
Collect five-node inference trace	_____	_____	_____	_____
Complete benchmark CSV	_____	_____	_____	_____
Write novelty comparison	_____	_____	_____	_____
Draft U.S. impact narrative	_____	_____	_____	_____
Counsel review of NIW and F-1 issues	_____	_____	_____	_____

11 | Final Deliverable Checklist

[ ] Architecture specification and sequence diagram

[ ] Versioned source code and reproducibility guide

[ ] Per-node execution logs and model partition trace

[ ] Raw benchmark dataset, scripts, charts, and negative results

[ ] Security and reliability test report

[ ] Related-work and novelty comparison

[ ] Technical research paper draft

[ ] Three-year research and deployment roadmap

[ ] Independent validation / pilot outreach evidence

[ ] Petitioner contribution and credential exhibit index

[ ] Attorney-reviewed proposed endeavor and NIW argument

12 | Suggested Petition Narrative - Working Draft

Proposed endeavor: to advance decentralized, resource-aware distributed AI inference systems that coordinate heterogeneous computing resources to expand access to capable AI workloads for resource-constrained U.S. researchers and organizations. RACoN is the principal experimental vehicle for investigating adaptive model partitioning, resource-aware placement, network overhead, reliability, and reproducible performance. Specific performance and impact claims will be limited to validated results.

Legal note: EB-2 eligibility must also be established separately (advanced degree or exceptional ability). This plan is not legal advice and does not predict approval. Review petition language and F-1 activity implications with qualified U.S. immigration counsel.